Tab 2

Task1_Documentation.pdf

INTERN & PROJECT DETAILS

COMPANY NAME. : CODTECH IT SOLUTIONS

INTERN NAME. : MALLAVARAPU LAVANYA

INTERN ID. : CITS1289

DOMAIN. : VLSI DESIGN

BATCH DURATION. : MAY-JUNE 2026

TASK TITLE：7-SEGMENT DISPLAY DRIVER

PROJECT OVERVIEW

This project implements a 4-bit Binary Coded Decimal (BCD) to 7-segment display decoder in Verilog. It converts a 4-bit digital input (representing values from 0 to 9) into the corresponding 7-bit control signal needed to drive a standard 7-segment display using active-high logic.

Design Specifications

- Hardware Description Language: Verilog HDL
- Target Domain: VLSI Design / Digital Systems
- Logic Type: Active-High (1 turns segment ON, 0 turns segment OFF)
- Output Format: 7-bit bus ordered as [g f e d c b a]

Segment Mapping Truth Table

Input (Dec)	BCD Input [3:0]	Output [g f e d c b a]	Displayed Character

0	0000	0111111	0
1	0001	0000110	1
2	0010	1011011	2
3	0011	1001111	3
4	0100	1100110	4
5	0101	1101101	5
6	0110	1111101	6
7	0111	0000111	7
8	1000	1111111	8
9	1001	1101111	9
INVALID	1010 to 1111	0000000	Blank

SOURCE CODE

```
//
===============================
============================
```

// COMPANY : CODTECH IT SOLUTIONS
// INTERN. NAME : MALLAVARAPU LAVANYA
// INTERN ID. : CITS1289
// DOMAIN. : VLSI
//. TASK TITLE. : MAY-JUNE 2026
//.
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VERILOG DESIGN CODE

```
`timescale 1ns / 1ps

module seven_seg_driver (
    input wire [3:0] bcd_in,      // 4-bit BCD input (0 to 9)
    output reg [6:0] seg_out      // 7-bit output for segments
(g f e d c b a)
);

    // Combinational logic block for BCD to 7-Segment
conversion
    always @(*) begin
        case (bcd_in)
            // Logic: 1 = Segment ON, 0 = Segment OFF
            // Mapping: seg_out = {g, f, e, d, c, b, a}
```

```
            4'b0000: seg_out = 7'b0111111; // Displays 0
            4'b0001: seg_out = 7'b0000110; // Displays 1
            4'b0010: seg_out = 7'b1011011; // Displays 2
            4'b0011: seg_out = 7'b1001111; // Displays 3
            4'b0100: seg_out = 7'b1100110; // Displays 4
            4'b0101: seg_out = 7'b1101101; // Displays 5
            4'b0110: seg_out = 7'b1111101; // Displays 6
            4'b0111: seg_out = 7'b0000111; // Displays 7
            4'b1000: seg_out = 7'b1111111; // Displays 8
            4'b1001: seg_out = 7'b1101111; // Displays 9
            default: seg_out = 7'b0000000; // Turns all
segments OFF for invalid BCD
        endcase
    end

endmodule
```

TEST BENCH CODE

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======================================
// COMPANY.   : CODE IT SOLUTIONS
//. INTERN NAME. : MALLAVARAPU LAVANYA
//. INTERN ID.   : CITS1289
//. DOMAIN.   : VLSI DESIGN
//. TASK TITLE.   : 7-SEGMENT DISPLAY DRIVER
//
===
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TEST BENCH CODE

```
`timescale 1ns / 1ps

module tb_seven_seg_driver;

    // Inputs to the Design Under Test (DUT)
    reg [3:0] bcd_in;

    // Outputs from the DUT
    wire [6:0] seg_out;

    // Instantiate the Unit Under Test (UUT)
    seven_seg_driver uut (
        .bcd_in(bcd_in),
        .seg_out(seg_out)
```

```
    );

    // Stimulus generation block
    initial begin
        // Setup value dumping for waveform generation
        $dumpfile("dump.vcd");
        $dumpvars(0, tb_seven_seg_driver);

        // Monitor system changes on the console terminal
        $monitor("Time = %0t ns | BCD Input = %b (%d) |
7-Seg Output (g-f-e-d-c-b-a) = %b",
                 $time, bcd_in, bcd_in, seg_out);

        // Test Case 0: Initialize
        bcd_in = 4'b0000; #10;

        // Test Cases 1 to 9: Loop through all valid BCD
values
        bcd_in = 4'b0001; #10;
        bcd_in = 4'b0010; #10;
        bcd_in = 4'b0011; #10;
        bcd_in = 4'b0100; #10;
        bcd_in = 4'b0101; #10;
        bcd_in = 4'b0110; #10;
        bcd_in = 4'b0111; #10;
        bcd_in = 4'b1000; #10;
        bcd_in = 4'b1001; #10;
```

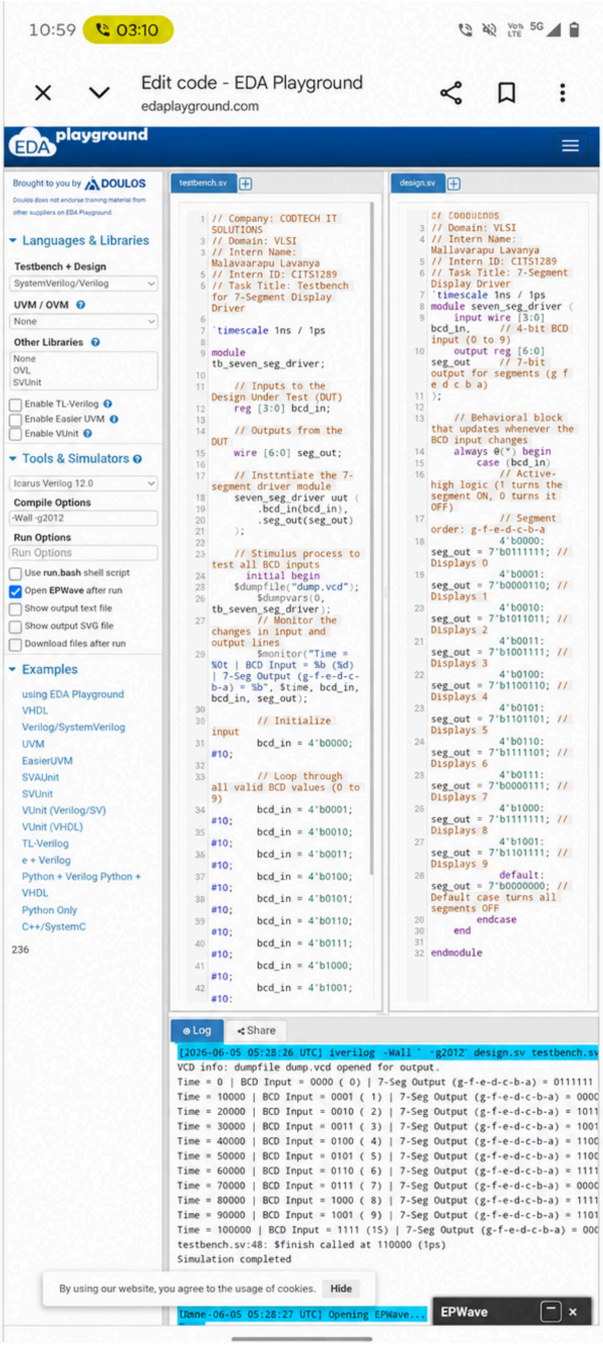
```
        // Test Case 10: Verify invalid BCD input handling (Default Case)
        bcd_in = 4'b1111; #10;

        // Terminate the simulation run cleanly
        $finish;
    end

endmodule
```

Simulation & Verification Results

Expected Console Output

When you compile and run this code on an EDA simulator (like EDA Playground, ModelSim, or Vivado), the console output will look exactly like this


10:59 02:49
Edit code - EDA Playground
edaplayground.com
EDA playground
Log
Share
[2026-06-05 05:28:26 UTC] iverilog -Wall ' -g2012' design.sv testbench.sv
VCD info: dumpfile dump.vcd opened for output.
Time = 0 | BCD Input = 0000 (0) | 7-Seg Output (g-f-e-d-c-b-a) = 0111111
Time = 10000 | BCD Input = 0001 (1) | 7-Seg Output (g-f-e-d-c-b-a) = 0000
Time = 20000 | BCD Input = 0010 (2) | 7-Seg Output (g-f-e-d-c-b-a) = 1011
Time = 30000 | BCD Input = 0011 (3) | 7-Seg Output (g-f-e-d-c-b-a) = 1001
Time = 40000 | BCD Input = 0100 (4) | 7-Seg Output (g-f-e-d-c-b-a) = 1100
Time = 50000 | BCD Input = 0101 (5) | 7-Seg Output (g-f-e-d-c-b-a) = 1101
Time = 60000 | BCD Input = 0110 (6) | 7-Seg Output (g-f-e-d-c-b-a) = 1111
Time = 70000 | BCD Input = 0111 (7) | 7-Seg Output (g-f-e-d-c-b-a) = 0000
Time = 80000 | BCD Input = 1000 (8) | 7-Seg Output (g-f-e-d-c-b-a) = 1111
Time = 90000 | BCD Input = 1001 (9) | 7-Seg Output (g-f-e-d-c-b-a) = 1101
Time = 100000 | BCD Input = 1111 (15) | 7-Seg Output (g-f-e-d-c-b-a) = 000
testbench.sv:48: $finish called at 110000 (1ps)
By using our website, you agree to the usage of cookies. Hide
[2026-06-05 05:28:27 UTC] Opening EPWave...
EPWave


10:59 03:10
Edit code - EDA Playground
edaplayground.com
EDA playground
Log
Share
[2026-06-05 05:28:26 UTC] iverilog -Wall ' -g2012' design.sv testbench.sv
VCD info: dumpfile dump.vcd opened for output.
Time = 0 | BCD Input = 0000 (0) | 7-Seg Output (g-f-e-d-c-b-a) = 0111111
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Time = 20000 | BCD Input = 0010 (2) | 7-Seg Output (g-f-e-d-c-b-a) = 1011
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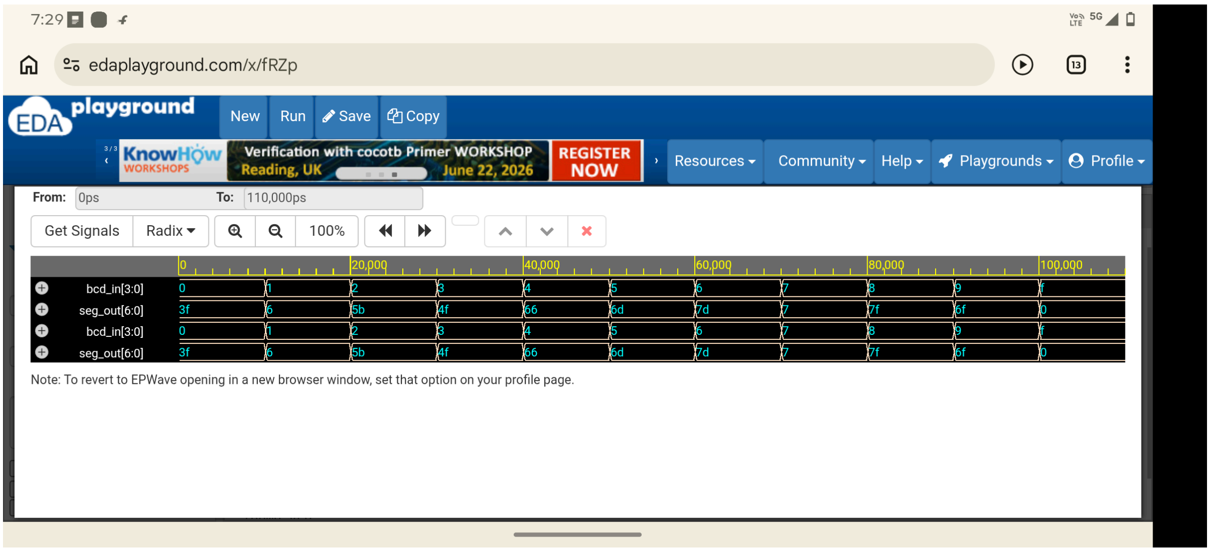
Time = 80000 | BCD Input = 1000 (8) | 7-Seg Output (g-f-e-d-c-b-a) = 1111
Time = 90000 | BCD Input = 1001 (9) | 7-Seg Output (g-f-e-d-c-b-a) = 1101
Time = 100000 | BCD Input = 1111 (15) | 7-Seg Output (g-f-e-d-c-b-a) = 000
testbench.sv:48: $finish called at 110000 (1ps)
Simulation completed
By using our website, you agree to the usage of cookies. Hide
Opening EPWave...
EPWave

TEXT :


7:29
edaplayground.com/x/fRZp
EDA playground
New
Run
Save
Copy
KnowHow WORKSHOPS
Verification with cocotb Primer WORKSHOP
Reading, UK
June 22, 2026
REGISTER NOW
Resources
Community
Help
Playgrounds
Profile
From: 0ps
To: 110,000ps
Get Signals
Radix
100%
0
20,000
40,000
60,000
80,000
100,000
bcd_in[3:0]
seg_out[6:0]
bcd_in[3:0]
seg_out[6:0]
0 1 2 3 4 5 6 7 8 9 f
3f 6 5b 4f 66 6d 7d 7 7f 6f 0
0 1 2 3 4 5 6 7 8 9 f
3f 6 5b 4f 66 6d 7d 7 7f 6f 0
Note: To revert to EPWave opening in a new browser window, set that option on your profile page.

CONCLUSION:

The design successfully translates BCD values into appropriate signals for a 7-segment display. The testbench verifies all operations, including proper blanking when encountering invalid inputs outside the 0–9 range.If you need help generating a formal project report or need to convert this code to an active-low configuration for a common anode display, let me know!

Tab 3